

Course syllabus for

Academic year 2022/2023 >

FFY012 - Solid state physics

Fasta tillståndets fysik

Course syllabus adopted 2021-02-08 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

7,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Chemical Engineering with Engineering Physics, Engineering Physics

Department: 16 - PHYSICS

Teaching language: Swedish

Application code: 57132

Open for exchange students: No

Block schedule: [C+](#)

Maximum participants: 185

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0113 Laboratory 1,5 c Grading: UG			1,5 c				
0213 Examination 6,0 c Grading: TH			6,0 c				Thu 16/03-2023 pm J, Fri 09/06-2023 pm J, Thu 17/08-2023 pm J

In programs

[TKKEF CHEMICAL ENGINEERING WITH ENGINEERING PHYSICS, Year 3 \(compulsory\)](#)

[TKTFY ENGINEERING PHYSICS, Year 3 \(compulsory\)](#)

Examiner:

[Eva Olsson](#)



[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Course in quantum physics

Aim

Solid state physics is an obvious part in the education in physics. The course is required for many later courses in the field.

Learning outcomes (after completion of the course the student should be able to)

- Mathematically describe crystal structures in terms of a Bravais lattice with a base; unit cell, primitive cell. Crystal planes, Miller Indices.
- Describe the most basic models of crystal binding; covalent bond, ionic bond, metallic bond, hydrogen bond.
- Describe and calculate how crystal structure can be determined by diffraction, and know the difference between x-ray, neutron, and electron diffraction. Account for the production of X-rays and neutrons, and the function of synchrotron light sources and neutron sources.
- Calculate the reciprocal lattice of a crystal structure factor for various types of structures, and explain the

concept of the Brillouin zone.

- Calculate the vibrational modes in crystals in the simple spring-ball model. Describe how quantised vibrations (phonons) contribute to the heat capacity and thermal conductivity. Be able to explain the difference between the acoustic and optical phonons.
- Describe and calculate the basic aspects of the free electron gas as given by the Fermi-Dirac distribution of a particle in a box or with periodic boundary conditions.
- Explain the concepts Fermi sphere, Fermi surface, Fermi wave vector, Fermi energy, Fermi temperature.
- Calculate the density of states depending on energy spectra and dimensionality.
- Describe the effect of electromagnetic fields through the Drude model for the complex conductivity. How this is related to the DC conductivity, reflectivity, refractive index, and plasma oscillations. Be able to give a simple explanation for the color of different metals.
- Describe the basic difference between metals and semiconductors / insulators using the band structure.
- Explain the meaning of the Bloch theorem for electrons in a periodic potential, and the term crystal momentum.
- Derive the band structure in a weak periodic potential from the empty lattice model, and using the tight-binding model for simple lattices.
- Describe and use the equation of motion of a Bloch electron and how this is related to the concept of effective mass.
- Describe the basic physics of a semiconductor, with direct or indirect band gap, intrinsic or doped. Conduction and valence band, and the description of the electrons in the valence band in terms of holes. Effective mass of bands, mobility concept, as well as the exponential temperature dependence of conductivity.
- Explain how to calculate the chemical potential and the electron / hole density for intrinsic or doped semiconductor.
- Describe the Hall effect and how this is related to the type of charge carriers.
- Describe how the Fermi surface is related to the band structure, and qualitatively derive the Fermi surface of a weak periodic potential.
- Describe the electrons eigenstate in a magnetic field in terms of Landau levels, and how oscillations in the physical properties can occur as a function of $1/B$.
- Give an elementary description of materials' magnetic properties: diamagnetism and (Curie) paramagnetism in insulators, Pauli paramagnetism and Landau diamagnetism of metals. Ferromagnetism as an effect of electron-electron interactions, and mean field theory of this. Magnons / spin wave excitations in a magnetically ordered state.
- Give an elementary description of a superconductor as a macroscopic quantum state. Describe the Meissner effect, and magnetic flux quantisation in a superconducting ring.

Content

The course provides an overview of the physical properties of solids, experimental methods used to explore them and how properties are explained on the basis of theoretical models at a microscopic level.

Initially it is described how the atoms are arranged in crystalline substances and how the order can be determined by diffraction of incident radiation (x-rays, electrons, neutrons) or via direct imaging methods. In the description of diffraction the reciprocal lattice is introduced, as an essential concept for the understanding of many of the properties of crystalline substances.

The following section describes vibrational waves and thermal properties derived from these (heat capacity, thermal conductivity). Then defects in the atomic arrangements are treated and their impact on various properties.

The course continues to discuss electronic properties (conductivity, optical reflectivity, plasma oscillations, Landau levels, the Hall voltage), starting first from the free electron model and thereafter, starting from a description of an electron in a periodic potential (energy gap, operating in the field, optical excitations, effective mass, holes). An important application is to intrinsic and doped semiconductors.

An overview of the magnetic properties of solids is given (diamagnetism, paramagnetism, ferromagnetism, the spin waves, domains). Finally, a brief introduction to superconductivity is given.

Organisation

Lectures, class room exercises, 4 laboratory exercises, "dugga", written exam

Literature

C Kittel: "Introduction to Solid State Physics", 8th (John Wiley and Sons, 2005)

P. Hofmann, Solid State Physics, An introduction. 2nd edition (Wiley-VCH, 2008)

Examination including compulsory elements

The course ends with a written exam comprising numerical and descriptive problems. There is a non-compulsory written test that may give bonus points to the exam. The course has four mandatory lab-exercises.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.